A Comparative Study of Kolmogorov-Arnold Network Models for Robust WiFi Fingerprinting-based Positioning

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***Abstract*— Indoor positioning has become an attractive research topic as people spend more time indoors. Determining the user's position in buildings is critical for safety applications, such as emergency response during fires or earthquakes. Among various approaches, WiFi Fingerprinting is a widely applied technique in WiFi-based indoor positioning systems. However, its performance often suffers from environmental variability. To address this challenge, this study presents a comprehensive evaluation of various Kolmogorov-Arnold Networks (KAN) models applied to WiFi Fingerprinting, which aims to reduce positioning errors influenced by changes in the environment. To the best of our knowledge, this study is the first comprehensive evaluation of KAN-based models for indoor positioning. The five models are tested in three public datasets, including UJIIndoorLoc, HCXY, and SYL. Experimental results demonstrate that the BSRBF-KAN model outperforms the others with the lowest positioning errors of 8.43 m, 2.18 m, and 4.96 m on the three datasets, respectively. Even after applying Random Forest to remove less valuable WiFi features, BSRBF-KAN still achieves superior performance. The findings reveal the strong potential of KAN-based approaches for improving indoor positioning accuracy and provide suggestions for model selection in practical deployments.**

Keywords—Indoor Positioning System, WiFi Signal, WiFi Fingerprinting, Kolmogorov-Arnold Network, Deep Learning

I. INTRODUCTION

As one of the key parts of location-based services, indoor positioning plays a crucial role in complex indoor environments, including hospitals, airports, shopping malls, etc. Outdoor positioning systems commonly use the Global Navigation Satellite System (GNSS). Nevertheless, applying the same approach indoors faces difficulties because GNSS signals are weakened or distorted by walls, ceilings, and other obstacles. These factors cause signal attenuation, reflections, and the absence of a direct satellite connection, which makes it hard to accurately determine user/asset position inside buildings. WiFi Fingerprinting, among various indoor positioning techniques, utilizes the pervasive setup of WiFi access points (APs) in modern buildings [1, 2]. Thus, it has become a promising and practical solution that utilizes existing infrastructure without additional hardware. The technique involves two main phases: an offline phase, where a site survey is performed to gather received signal strength (RSS) values at known coordinates, so-called reference points (RPs), resulting in an offline fingerprint database, and an online phase, where an unknown position is estimated by comparing the new set of RSS values to the database.

Despite its practicality, WiFi Fingerprinting-based positioning systems are still experiencing constraints such as the high dimensionality of WiFi data, the sparse AP availability, and the temporal instability caused by environmental changes, user mobility, and interference. Traditional approaches, like deterministic and probabilistic, have shown limited ability to capture the nonlinear relationships between RSS patterns and physical coordinates. In contrast, deep learning based-approaches have been utilized to improve the positioning accuracy by learning complex RSS value-position relationships.

Recently, Kolmogorov-Arnold Network (KAN) [3], a new type of neural architecture inspired by the Kolmogorov-Arnold representation theorem, has been introduced to the research community. According to the theorem, any continuous function with multiple variables can be represented as a finite combination of continuous univariate functions connected through addition. Based on this concept, rather than relying on static weighted edges as in multilayer perceptrons (MLPs), KAN utilizes learnable univariate activation functions, often implemented as B-splines, along the connections. This approach moves the learning process from fixed weights to flexible, data-driven mappings between variables. As a result, KAN provides strong interpretability, efficient use of parameters, and enhanced modeling flexibility in function approximation, making it particularly effective for tasks with low-dimensional input spaces.

Up to now, various versions of KAN have been proposed to adapt the architecture to different application domains [4, 5]. Each model involves a trade-off among accuracy, computational efficiency, resistance to noise, and flexibility in handling heterogeneous data distributions. For the WiFi Fingerprinting-based positioning systems, KAN-based models have become potential solutions since the per-edge

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learnable activation functions enable the modeling of WiFi signal fluctuations caused by obstacles, multipath effects, and environmental variability. Some KAN-based models can adapt to spatial heterogeneity, e.g., differences in signal propagation between floors or buildings, due to the ability to integrate different basis functions [6] or dynamically select function spaces [7]. In addition, some models have been optimized for computational efficiency, which allows them to perform in real time even on resource-constrained platforms like mobile devices [8, 9].

Nonetheless, a comparative evaluation of KAN-based models for WiFi Fingerprinting-based positioning systems is still lacking, particularly in determining which model achieves the lowest positioning errors in diverse and complicated datasets. In this paper, motivated by architectural diversity and practical suitability for positioning tasks, we conduct a comparative empirical evaluation of five KAN-based models applied in the WiFi Fingerprinting technique, including original KAN, FastKAN [8], BSRBF-KAN [6], S-KAN [7], and FC-KAN [10]. The key contributions of this work are summarized as follows.

- We implement the five KAN-based models, which are introduced to address different aspects of the original KAN's limitations.
- We evaluate the five models in three public datasets, including UJIIndoorLoc, HCXY, and SYL, which were collected in varied and complex real-world settings.
- Based on empirical results and theoretical analysis, we identify the most effective and robust model that achieves the lowest positioning error under varying indoor environments.

The remainder of this paper is structured as follows: Section 2 provides a review of related studies on WiFi fingerprinting and KAN-based models applied to positioning. Section 3 details the implemented KAN-based models that are used for evaluation. Section 4 presents and discusses the experimental results, while Section 5 concludes the study and offers directions for future work.

II. Related Work

A. WiFi Fingerprinting-based Positioning Systems

Over the last two decades, WiFi Fingerprinting has been extensively researched as a highly promising indoor positioning technique. Two conventional approaches, deterministic and probabilistic, are commonly applied in this technique to determine the user's position. The first approach, with the representatives as the nearest neighbors-based methods [11, 12], offers speed and simplicity, making them attractive for lightweight deployments. Even so, this approach often lacks robustness in noisy and dynamic environments. The second approach, even though it provides better modeling of uncertainty and can obtain higher positioning accuracy under variable conditions [13, 14], requires more extensive data collection and computational resources.

Recently, with the rise of deep learning, neural networks have been increasingly applied to WiFi Fingerprinting [15, 16]. MLPs, by learning nonlinear feature transformations, showed promising positioning results [17, 18]. Meanwhile, Recurrent Neural Networks attempted to model temporal dependencies in complicated positioning areas, especially when the user was walking [19, 20]. In addition, Convolutional Neural Networks, the technique that dealt with grid-like data, treated RSS maps as 2D images to capture spatial correlations between APs [21, 22]. Nevertheless, these models use fixed activation functions and linear connections, which may not efficiently capture the complex characteristics in WiFi data correlated to physical RPs.

B. KAN-based Models in Indoor Positioning

The above trade-off leads to the need to combine the nonlinear function approximation capability of neural networks with the statistical adaptability of probabilistic reasoning, which is presented in KAN. Although recently introduced, KANs have been applied in various research topics, such as intrusion detection [23, 24], image classification [25, 26], human activity recognition [27], etc. For indoor positioning, KAN-based models have also been applied to enhance the tracking accuracy. In [28], by utilizing spatial magnetic field characteristics, Gao et al. proposed the MP-KAN algorithm to reduce the error in near-field predictions and balance multi-axis errors. From the experimental results, the maximum error was only 0.24 mm. Moreover, KAN was also applied in a Visible Light-based positioning system, which also demonstrated sub-centimeter accuracy, even with tilted light-emitting diodes [29]. In [30], Wang et al. introduced the CCKAN, which integrated the complex convolutional networks with KAN, to improve Angle of Arrival estimation in near-field scenarios, where far-field assumptions broke down and affected the accuracy. The simulation demonstrated that CCKAN consistently surpassed conventional algorithms in various conditions.

For WiFi-based positioning, only a few published studies have used KAN-based models to improve the positioning accuracy. Yan et al. [31] proposed RSLoc with the KAN convolution network to generate stable fingerprint mappings, which incorporated some optimizations, such as removing invalid APs and applying entropy regularization. The positioning results surpassed those of other methods, achieving up to a 48% improvement in accuracy after eliminating invalid APs. Furthermore, in [32], Krisnawan et al. introduced a hybrid RBF-KAN model for accurate indoor positioning using dense grid WiFi Fingerprinting. The model attempted to handle non-linearities and adapt to high-dimensional functions. Compared to other methods, the proposed model achieved a mean positioning error of only 0.105 m, with a positioning accuracy of 98.26%.

It can be seen that KAN-based models are suitable for WiFi Fingerprinting-based positioning systems due to learnable univariate functions on edges for modeling highly nonlinear, location-specific RSS values-position relationships. In contrast, the positioning performance of different KAN-based models over diverse datasets remains uncertain. Therefore, this work implements and evaluates the performance of various KAN-based models to find the best model that achieves the lowest positioning errors while maintaining a reasonable computational cost.

III. KAN-based Models

As shown in Fig. 1, all the KAN-based models are built upon the Kolmogorov-Arnold representation theorem. The original KAN is the first model to implement the above concept in neural networks. Its most unique feature is the replacement of static weights with learnable one-dimensional functions, i.e., B-splines, which allows each edge to act as a

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nonlinear, adaptive transformation. This model opens a high level of interpretability, however, one major limitation of the original KAN is its high computational cost caused by the use of B-splines and the requirement for grid rescaling during training.

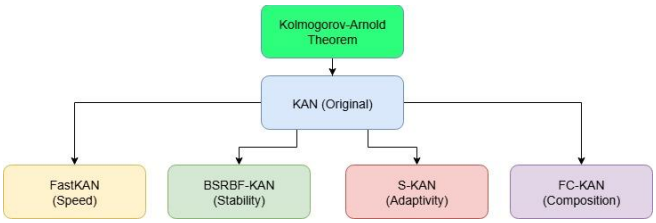


Fig. 1. Development of KAN-based models

Since the introduction of the original KAN, several KAN-based models have been proposed to cover its limitations, such as high computational complexity, excessive memory usage, limited domain adaptability, scalability constraints, etc. Therefore, this study considers four KAN-based models to handle the aforementioned issues, as presented in Fig. 1: FastKAN, which enhances computational speed and efficiency; BSRBF-KAN, which improves stability and generalization; S-KAN, which increases adaptivity; and FC-KAN, which augments compositional feature learning.

FastKAN is proposed to reduce the high computational cost of the original KAN by approximating its B-spline basis functions with Gaussian Radial Basis Functions (RBFs). This simplification converts the model into a form of RBF networks, offering lower computational demands and simpler implementation. FastKAN speeds up forward-pass computations by over three times through the removal of de Boor-Cox iterations and adaptive grid rescaling.

Meanwhile, BSRBF-KAN is proposed to enhance model stability and learning convergence. The model linearly combines B-splines and RBFs at each edge, which utilizes the smoothness and local control of splines together with the adaptability and resilience of RBFs. Thus, it maintains reliable performance while significantly reducing computational demands.

Next, S-KAN is suggested to allow each node to dynamically select the most appropriate activation function space during training, instead of assigning one activation function per edge or node. In practical terms, the model offers a significant improvement in generalization and task-specific performance. Nevertheless, S-KAN introduces substantial complexity, requiring extra selection mechanisms, possible use of multi-objective loss functions, and higher computational demands during both training and inference.

Lastly, FC-KAN generalizes the KAN model by incorporating multiple basis functions and mathematical operations at the output of each edge or node. Specifically, it supports various function combinations, such as element-wise addition, multiplication, quadratic transformations, Difference of Gaussians (DoG), etc. These operations are mainly applied in low-dimensional layers to manage computational costs. FC-KAN aims to enhance feature richness and pattern recognition by modeling higher-order interactions among input features. Nevertheless, the limitation of this model is the computational complexity. The main characteristics of the five KAN-based models are summarized in Table I.

TABLE I. COMPARISON OF KAN-BASED MODELS

Models	Characteristics
KAN	Learnable activation functions on edges (splines), no linear weights
FastKAN	Speed up KAN by replacing B-splines with Gaussian RBFs
BSRBF-KAN	Combine B-splines with RBFs in activation functions to improve stability and convergence
S-KAN	Nodes adaptively select the best activation function during learning
FC-KAN	Use various combinations of functions (sum, product, concat) to capture deeper representations

Among the five models, KAN is the simplest one since it is the most basic model. In contrast, FC-KAN is the most complex model since it incorporates multiple mathematical function combinations in low-dimensional layers, requiring careful handling of tensor operations and computational overhead. FastKAN is a bit more complicated than KAN by replacing B-splines with computationally leaner RBFs, followed by BSRBF-KAN, since this model merges B-splines and RBFs. Lastly, S-KAN is simpler than FC-KAN, which introduces per-node adaptivity, requiring dynamic selection and additional control logic.

Fig. 2 illustrates the application of KAN-based models to WiFi Fingerprinting to determine a user's position. The input layer involves the nodes, which are RSS values, i.e., fingerprints, collected from the available APs at different RPs. Each connection between nodes is substituted with a trainable univariate function ϕ, which allows flexible nonlinear transformations. The function's equation differs among the various KAN-based models, as detailed in the respective references. These functions are structured hierarchically across KAN layers, with multiple transformations composed to sequentially model complex RSS value-position relationships. The green circular nodes present the number of units after each transformation stage, e.g., 512, 256. In the output layer, the transformed features are projected onto the user's 2D position coordinates.

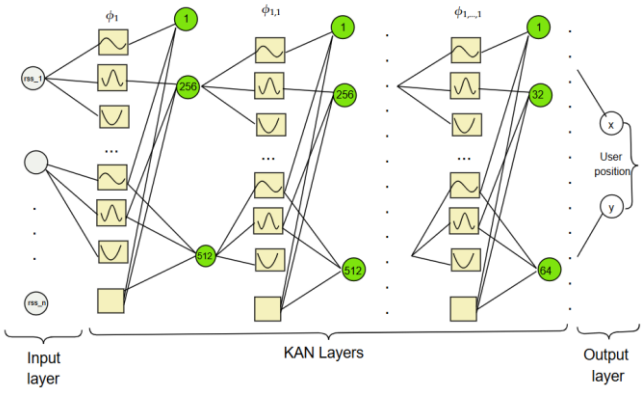


Fig. 2. Structure of the KAN-based models applied to WiFi Fingerprinting

IV. EXPERIMENTAL RESULTS

A. Datasets

In this work, three WiFi Fingerprinting public datasets are used for performance evaluation: UJIIndoorLoc [33], HCXY [34], and SYL [34]. The first dataset was collected within three different buildings spanning multiple floors at Jaume I University (Spain), while the two other datasets were collected from two buildings' floors in Xuzhou and Jinan cities (China). These datasets were collected with the participation of many

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volunteers as well as many smartphones. Each dataset includes the training and testing files, but the number of WiFi data collected for each file is different. The more specific characteristics for the three datasets are presented in Table II. It should be noted that, in this paper, since we do not focus on building classification, the UJIIndoorLoc dataset is split into three sub-datasets corresponding to three buildings.

TABLE II. MAIN CHARACTERISTICS OF THREE DATASETS

Characteristics	Datasets		
	UJIIndoorLoc	***HCXY***	***SYL***
Total areas (m^2)	108,703	3,600	2,600
Total number of RPs	735	379	296
Distance between adjacent RPs (m)	~3.65	1.2	1.2
Samples per RP	≥ 2	30	30
Total number of APs	520	347	363

B. Results

The positioning performance of the five KAN-based models is evaluated using the mean positioning error as the comparison metric. This metric is calculated as the average Euclidean distance between the actual test points and their estimated positions. The model parameters used in the experiments are listed in Table III.

TABLE III. THE PARAMETER SETTING FOR KAN-BASED MODELS

Parameters		Values
Hidden Layers		[512, 512, 256, 256, 128, 128, 64, 64]
Model	KAN	SplineLinear
	FastKAN	RBF, SplineLinear
	BSRBF-KAN	func_list=[SiLU, RBF], grid_size=4, spline_order=3
	S-KAN	func_list=[shifted_softplus]
	FC-KAN	func_list=['dog', ' B-splines'], grid_size=5, spline_order=3
Optimizer		Adam (Learning rate=0.001)
Batch Size		64

Fig. 3 to Fig. 5 display the positioning errors of the five KAN-based models on the three datasets UJIIndoorLoc, HCXY, and SYL, respectively. As shown in Fig. 3, the BSRBF-KAN model demonstrates the highest performance among the five models when evaluated on the UJIIndoorLoc dataset, with a positioning error of 8.43 m, which is lower than the others by 24.59% to 46.91%. These results demonstrate the effectiveness of BSRBF-KAN in WiFi Fingerprinting. On the UJIIndoorLoc dataset, previously published studies applying deep neural networks [35, 36] or CNNs [37, 38] reported positioning errors ranging from 9.29 m to 14.93 m. In comparison, BSRBF-KAN achieves lower errors, which represents an improvement of over 9.26% relative to these methods on the same dataset. This result can come from the hybrid integration of B-splines and RBFs, which strengthens the network's capacity to capture both local and global WiFi signal patterns, resulting in improved robustness to environmental changes. S-KAN is the runner-up model with an error of 11.18 m, followed by FastKAN with slightly higher error. In contrast, the highest positioning error belongs to the original KAN, with an error of approximately 16 m, which proves that this model lacks optimizations for dealing with the highly fluctuating RSS values. Furthermore, FC-KAN also performs poorly with an error of 15.18 m, showing that its compositional function approach may not be suitable for the WiFi Fingerprinting technique.

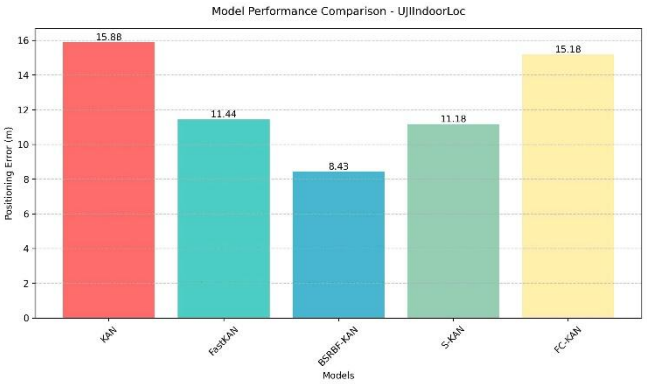


Fig. 3. Mean positioning errors of five models for the UJIIndoorLoc dataset

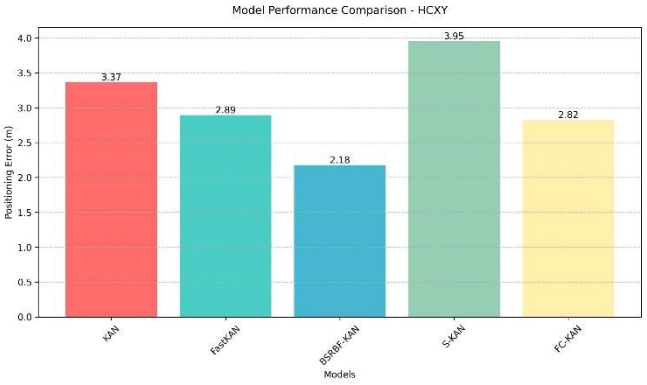


Fig. 4. Mean positioning errors of five models for the HCXY dataset

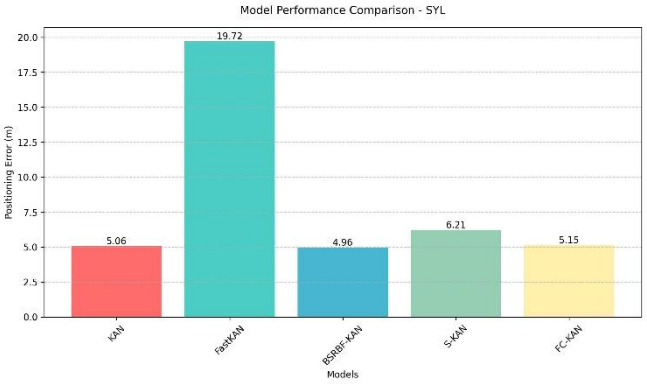


Fig. 5. Mean positioning errors of five models for the SYL dataset

For Fig. 4 and Fig. 5, even when testing the models on the two datasets HCXY and SYL, the BSRBF-KAN still achieves the best performance with the lowest positioning error of 2.18 m and 4.96 m. The results confirm its strong adaptability over various datasets due to its hybrid B-spline and RBFs. Conversely, the worst results in the HCXY and SYL datasets fall under the S-KAN and FastKAN, respectively. This indicates that the other KAN-based models do not consistently achieve the same performance over various datasets,

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excluding the BSRBF-KAN. The poor results of FastKAN on the SYL dataset reveal that prioritizing computational speed can reduce representational capability in noisy environments. Meanwhile, the adaptive mechanism of S-KAN may become unstable when AP distributions are highly variable.

Moreover, we apply the Random Forest (RF) algorithm, which is effective for feature selection and serves as an indirect form of dimensionality reduction, to the offline fingerprint database to identify the most significant APs. In the database, each AP plays a role as a feature. However, there exist APs that are either invisible in most locations or provide noisy signals. To address this issue, RF can compute feature importance to rank APs and remove the least informative ones. This may help improve positioning accuracy while reducing dimensionality, noise, and computational cost. Table IV shows the positioning errors of the five KAN-based models when applied to the dimensionally reduced fingerprint database by using the RF algorithm. As can be seen from the table, all models generally obtain lower positioning errors compared to the case without applying the RF algorithm, which demonstrates the effectiveness of RF for selecting meaningful features, i.e., APs. To be specific, the BSRBF-KAN is still superior to other KAN-based models since it achieves the lowest positioning errors with 8.03 m, 1.94 m, and 3.89 m in UJIIndoorLoc, HCXY, and SYL datasets, respectively. These results once again prove the high performance of the BSRBF-KAN model, even when considering the dimensionality reduction.

TABLE IV. MEAN POSITIONING ERRORS OF FIVE MODELS WHEN COMBINING WITH THE RF ALGORITHM

Models/Errors (m)	**Datasets**		
	UJIIndoorLoc	***HCXY***	***SYL***
KAN	15.49	3.16	4.67
FastKAN	10.79	2.56	7.74
BSRBF-KAN	**8.03**	**1.94**	**3.89**
S-KAN	10.65	3.56	5.48
FC-KAN	14.31	2.92	5.15

Overall, over the aforementioned results from different settings, BSRBF-KAN consistently ranks first with the lowest mean positioning error, hence, the model demonstrates its robustness when applied to WiFi Fingerprinting. The other models can achieve reasonable results in one dataset, but worse results in other ones, which suggests that those models may not generalize equally well in various scenarios without tuning. BSRBF-KAN may outperform other models due to superior robustness to noise and reduced overfitting. The model combines B-spline functions for precise local adaptability with RBFs for global smoothness, which enables the ability to capture both fine-grained RSS fluctuations and large-scale spatial patterns. Thus, it effectively models the nonlinear relationship between the WiFi signal and position. Nonetheless, accuracy is not the only factor that determines the suitability of a model for real-world deployment. BSRBF-KAN, although the most accurate, suffers from high computational cost, especially when compared to FastKAN, since it introduces dual basis computations that increase parameter count and training overhead. This illustrates the balance that needs to be maintained between accuracy and computational efficiency.

V. CONCLUSION

This study presented a comprehensive evaluation of five KAN-based models applied to WiFi Fingerprinting. Experiments on three public datasets demonstrated that KAN-based models can effectively deal with the nonlinear, environment-dependent characteristics of WiFi Fingerprinting-based indoor positioning systems. The results confirmed the idea that integrating hybrid basis functions into KAN architectures can lead to more stable and accurate positioning in diverse indoor environments. Among the five implemented KAN-based models, BSRBF-KAN was the best model since it consistently achieved the lowest positioning errors due to its hybrid basis function design that captured both local and global patterns in WiFi data. Future work may concentrate on implementing and testing the models in real-world applications. Moreover, the integration of WiFi with other technologies, such as Bluetooth, inertial measurement units, etc., should be investigated to further enhance positioning accuracy and stability.

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